

7.4.2 abr-summary (OSPF area)

Function

The **abr-summary** command configures route summarization on an area border router (ABR).

The **undo abr-summary** command disables route summarization on an ABR.

By default, route summarization is not configured on ABRs.

Product	Support
S6720EI, S6720S-EI, S5720HI, S5720EI, S6720SI, S6720S-SI, S5730SI, S5730S-EI, S5720SI, S5720S-SI, S6720LI, S6720S-LI, S5720LI, S5720S-LI, S2720EI, S1720X-E, S1720GW-E, S1720GWR-E, S1720GW, S1720GWR, S1720X	Supported
S5710-X-LI, S5700LI, S5700S-LI, S2750EI, S1720GFR	Not supported

Format

abr-summary *ip-address mask* [**cost** { *cost* | **inherit-minimum** }] [**advertise** [**generate-null0-route**] | **not-advertise** | **generate-null0-route** [**advertise**]]]
*

undo abr-summary *ip-address mask*

Parameters

Parameter	Description	Value
<i>ip-address</i>	Specifies the IP address of a summarized route.	The value is in dotted decimal notation.
<i>mask</i>	Specifies the mask of the IP address of the summarized route.	The value is in dotted decimal notation.
advertise not-advertise	Indicates whether to advertise the summarized route. By default, the summarized route is advertised.	-
cost <i>cost</i>	Specifies the cost of the summarized route. By default, the highest cost of specific routes is used as the cost of the summarized route.	The value is an integer that ranges from 0 to 16777214.

Parameter	Description	Value
inherit-minimum	Indicates that the smallest cost of specific routes is used as the cost of the summarized route.	-
generate-null0-route	Generates a blackhole route to prevent routing loops.	-

Views

OSPF area view

Default Level

2: Configuration level

Usage Guidelines

Usage Scenario

On a large-scale OSPF network, route search speed may decrease due to a large routing table size. Route summarization can be configured to reduce the size of the routing table and simplify management.

Route summarization aggregates multiple routes with the same IP prefix into one. If a link connected to a device within a summarized IP address range alternates between Up and Down states, the link status change is not advertised to the devices outside that IP address range. This prevents route flapping and improves network stability.

When the ABR sends routing information to other areas, it originates Type 3 LSAs for each network segment. If any contiguous segments exist in this area, run the **abr-summary** command to summarize these segments into one. The ABR then sends just one summarized LSA, and no LSAs that belong to the summarized network segment specified by the command. Therefore, the routing table size is reduced, and switch performance is improved.

Prerequisites

The **network** command has been run to specify the segments that need to be summarized before configuring route summarization.

Precautions

- This command applies only to ABRs for intra-area route summarization. The **asbr-summary** command configures AS Boundary Routers (ASBRs) to summarize the routes imported by OSPF.
- Route summarization cannot be configured on ABRs in different areas of the same process.

Example

```
# In OSPF 100 area 1, summarize routes in two network segments, 10.42.10.0 and 10.42.110.0, into one route 10.42.0.0, and advertise the summarized route to other areas.
```

```
<HUAWEI> system-view
[HUAWEI] ospf 100
[HUAWEI-ospf-100] area 1
[HUAWEI-ospf-100-area-0.0.0.1] network 10.42.10.0 0.0.0.255
[HUAWEI-ospf-100-area-0.0.0.1] network 10.42.110.0 0.0.0.255
[HUAWEI-ospf-100-area-0.0.0.1] abr-summary 10.42.0.0 255.255.0.0
```

Related Topics

[7.4.4 area \(OSPF\)](#)

[7.4.67 ospf](#)

7.4.3 advertise mpls-lsr-id

Function

The **advertise mpls-lsr-id** command configures OSPF to advertise MPLS LSR IDs to multiple areas as intra-area routes.

The **undo advertise mpls-lsr-id** command cancels the configuration.

By default, OSPF does not advertise MPLS LSR IDs to multiple areas as intra-area routes.

Format

advertise mpls-lsr-id [*cost cost*]

undo advertise mpls-lsr-id

Parameters

Parameter	Description	Value
cost <i>cost</i>	Specifies the cost of the advertised route.	The value is an integer that ranges from 0 to 65535. By default, it is 0.

Views

OSPF view

Usage Guidelines

Usage Scenario

The prerequisite of a valid tunnel is that an intra-area route to the egress is reachable. When an ABR serves as the egress of tunnels in two areas, OSPF considers that only one tunnel is valid, because there is only one intra-area route to the egress, namely the IP address of the ABR's loopback interface (used as an MPLS LSR ID for tunnel establishment). To the other areas, this route is an inter-area route. To allow the tunnels in both areas to take effect, run the **advertise mpls-lsr-id** command to configure OSPF to advertise the intra-area route to the MPLS LSR ID to all areas connected to this device.

Prerequisites

Before running this command, the **mpls te** command has been run to enable MPLS TE globally.

Configuration Impact

Running this command may have the following impacts:

- If the OSPF area where the local switch resides does not have the interface whose IP address is the MPLS LSR ID, the type of the OSPF routes (with the destination address being the MPLS LSR ID) on the other devices in this OSPF area will be changed from inter-area to intra-area, and the cost will also be changed.
- If an OSPF NSSA has the interface whose IP address is the MPLS LSR ID and the forwarding address of the NSSA LSAs advertised in this area is the MPLS LSR ID, the forwarding address of these NSSA LSAs will be changed and the NSSA LSAs will be re-advertised, thus causing route calculation.

Example

Configure OSPF to advertise MPLS LSR IDs to multiple areas as intra-area routes.

```
<HUAWEI> system-view
[HUAWEI] ospf
[HUAWEI-ospf-1] advertise mpls-lsr-id
```

7.4.4 area (OSPF)

Function

The **area** command creates an OSPF area and displays the OSPF area view.

The **undo area** command deletes a specified area.

By default, the system does not create any OSPF area.

Product	Support
S6720EI, S6720S-EI, S5720HI, S5720EI, S6720SI, S6720S-SI, S5730SI, S5730S-EI, S5720SI, S5720S-SI, S6720LI, S6720S-LI, S5720LI, S5720S-LI, S2720EI, S1720X-E, S1720GW-E, S1720GWR-E, S1720GW, S1720GWR, S1720X	Supported
S5710-X-LI, S5700LI, S5700S-LI, S2750EI, S1720GFR	Not supported

Format

area *area-id*

undo area *area-id*